

Advanced cryptography

Assignment-1: Is set of odd numbers with binary operations $(+)$, i.e., $\langle O, + \rangle$ an abelian group? If not explain the reasons with necessary notation.

Ans: No, the set of odd integers under addition is not an abelian group.

Let, $O = \{2k+1 \mid k \in \mathbb{Z}\}$ and consider the binary operation $+$. To be an abelian group $\langle O, + \rangle$ must satisfy the group axioms.

1. Closure

$a = 2m+1 \in O$ and $b = 2n+1 \in O$. Then

$$a + b = (2m+1) + (2n+1)$$

$$= 2(m+n+1)$$

which is even, so $a+b \notin O$ in general.

~~con-2~~ $1+3=4 \notin O \Rightarrow$ closure fails.

2. Associativity: Addition of integers is associative, so, this axiom would hold (inherited from $\mathbb{Z}$).

3. Identity element:

A group of identity e must satisfy $a + e = a$ for all $a \in \mathbb{O}$. In $\mathbb{Z}$ the additive identity is 0. but $0 \notin \mathbb{O}$ (0 is even).

Since there is no element of $\mathbb{O}$ that acts as an additive identity for all odd integers, no identity in $\mathbb{O}$.

4. Inverses: For $a = 2k+1 \in \mathbb{O}$, $\Rightarrow$

$$-a = -(2k+1)$$

$$= -2k-1$$

$$= 2(-k-1)+1 \in \mathbb{O}.$$

So, each element does have an additive inverse inside $\mathbb{O}$.

5. Commutativity: Addition is commutative,

so, if all other axioms held the group would be abelian.

Because closure and existence of an identity fail, $\langle \mathbb{O}, + \rangle$ is not a group of an abelian.